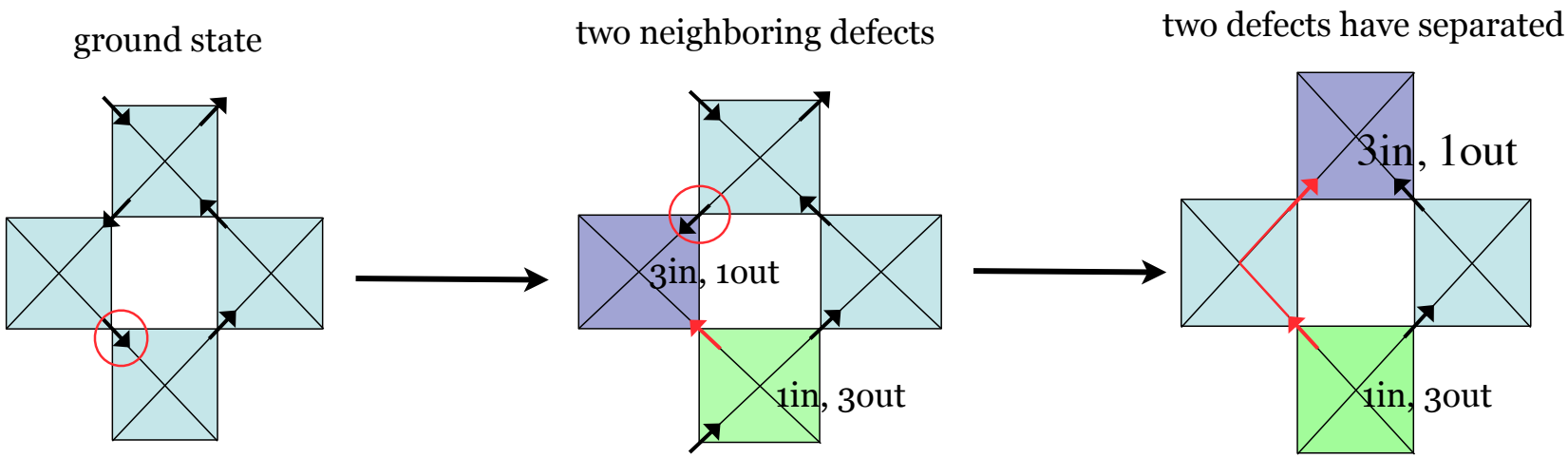
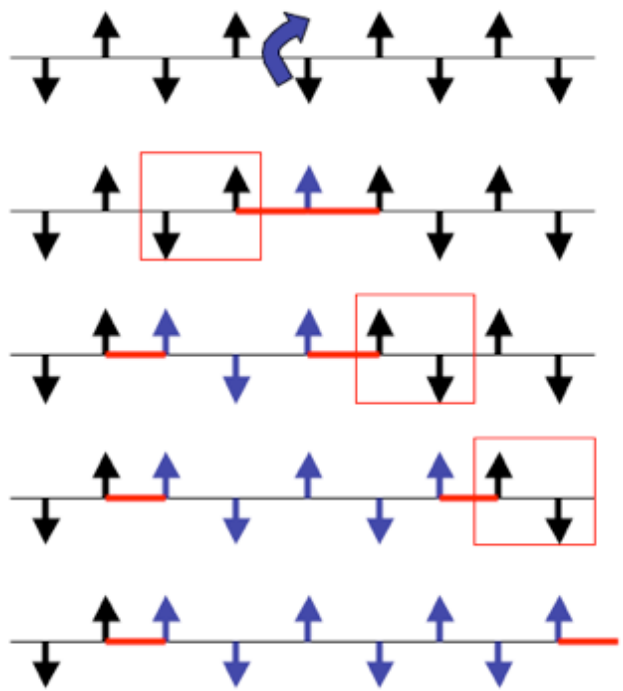


Spinons are natural in d=1

Bethe's solution: 1933
identification of spinons: 1981

S=1 spin wave breaks into two domain walls / spinons: hence each is carrying S=1/2



ground state
(only a cartoon)

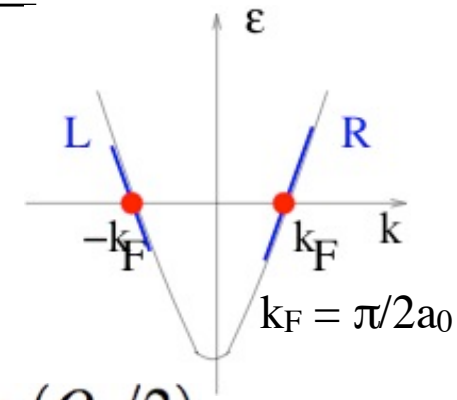
two neighboring defects
(domain walls)

domain walls have separated

domain walls have separated

Two-spinon continuum of spin-1/2 chain

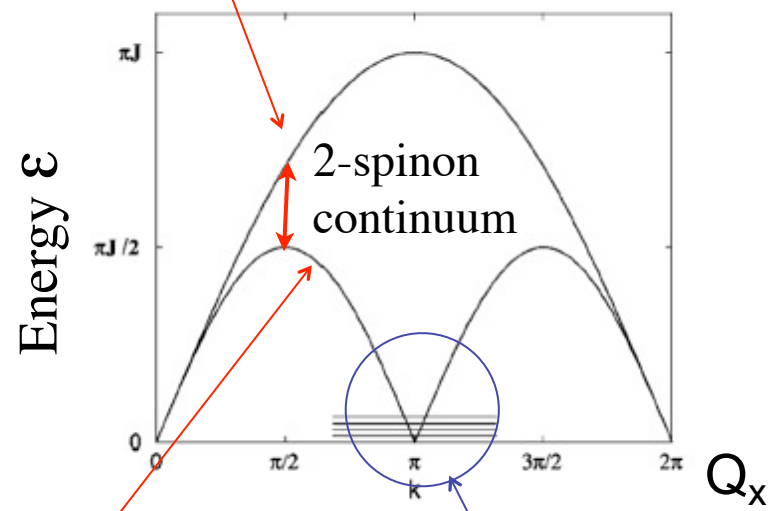
Spinon energy $\omega(k_x) = \frac{\pi J}{2}|\sin(k_x)|$ de Cloizeaux-Peason dispersion, 1962



S=1 excitation $\begin{cases} \varepsilon = \omega(k_{x1}) + \omega(k_{x2}) \\ Q_x = k_{x1} + k_{x2} \end{cases}$

$$\omega_{2,\text{upper}} = \pi J \sin(Q_x/2)$$
$$k_{x1} = k_{x2} = Q_x/2$$

Upper boundary



$$\varepsilon = \pi J \sin(\frac{Q_x}{2}) \cos(\frac{k_{x1} - k_{x2}}{2})$$

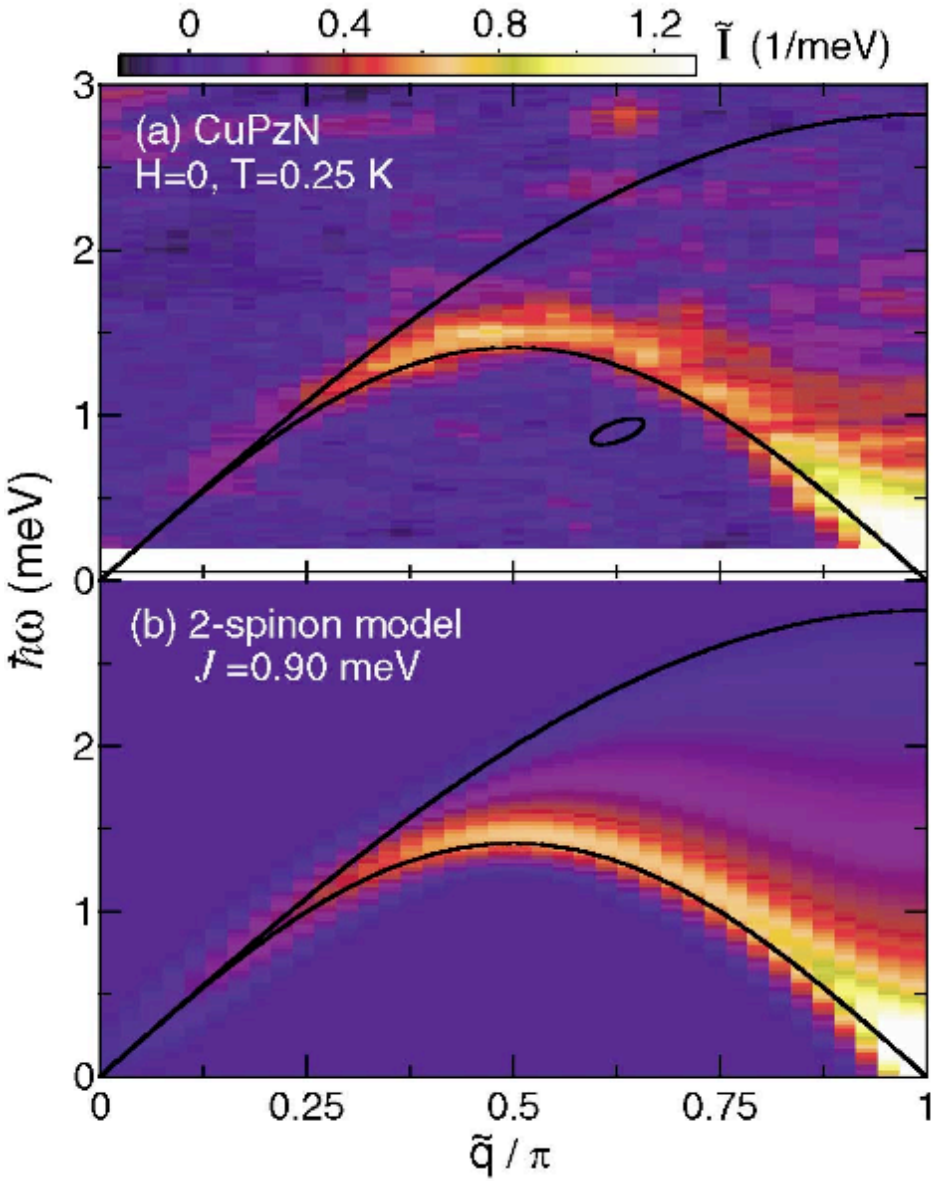
Variables:
k_{x1} and k_{x2}
or
ε and Q_x

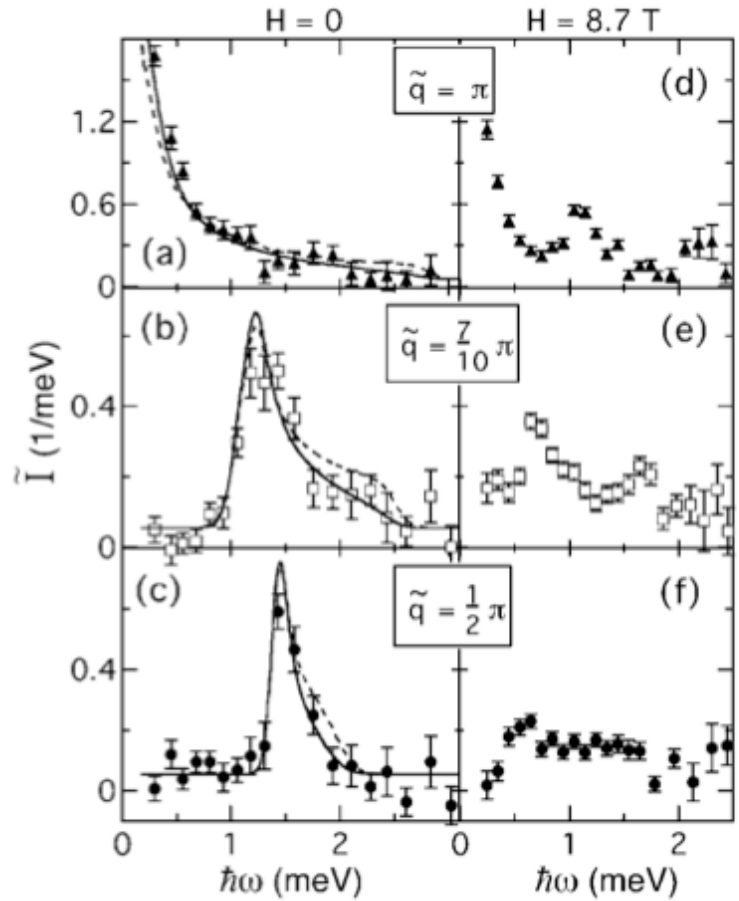
Lower boundary

Low-energy sector $Q_{AFM}=\pi$

$$\omega_{2,\text{lower}} = \frac{\pi J}{2} \sin(Q_x)$$
$$k_{x1} = 0, k_{x2} = Q_x$$

Highly 1d antiferromagnet CuPzN





Stone et al, PRL 2003

$J'/J < 10^{-4}$